

Short Test

Modern Code
2 (Methods & Arrays) - Reference Version
SS252025

Surname(s):

Given name(s):

Student ID / Mat.-Nr.:

	maximum	achieved
Exercise 1	5	
Exercise 2	5	
Exercise 3	2	
Total:	12	

Time allowed: 20 minutes

DO NOT TURN OVER UNTIL INSTRUCTED!

Instructions: For this test, you only need to write the requested Java code. You do **not** need to include solution ideas, test cases, or explanations - just the code itself.

Question 1 (5 points) Implement the method `processArray` that takes one parameter: a `String[]` `arr` (array of strings). The method should check if the array contains the string "mtd".

- If "mtd" is found in the array, reverse all strings in the array and return the modified array.
- If "mtd" is not found, convert all strings in the array to uppercase and return the modified array.

Example: If `arr = ["cat", "dog", "mtd", "rat"]`, the method should return `["tac", "god", "dtm", "tar"]` (since "mtd" is found, all strings are reversed).

If `arr = ["cat", "dog", "rat"]`, the method should return `["CAT", "DOG", "RAT"]` (since "mtd" is not found, all strings are converted to uppercase).

Question 2 (5 points) Write a method `countAnimals` that takes a `String[] animals` (array of animal strings) and returns a new array containing the total count for each animal type. Each string in the input array has the format "`<number> <animal>`" (e.g., "3 dogs", "5 cats"). The word is always the same (you don't need to handle plural forms or typos). Sum up the counts for each animal type and return an array of strings in the same format.

Example: If `animals = ["3 dogs", "5 cats", "8 dogs", "1 chicken", "2 chicken"]`, the method should return `["11 dogs", "5 cats", "3 chicken"]` (3+8=11 dogs, 5 cats, 1+2=3 chicken).

If `animals = ["2 dogs", "3 dogs"]`, the method should return `["5 dogs"]`.

Note: The order of animals in the output should match the order they first appear in the input array.

Question 3 (2 points) The following code has errors. Identify what is wrong and fix the code by adding or crossing out code directly in the code block below:

What's wrong:

```
static String[] reverseArray(String[] arr) {  
    String[] result = new String[arr.length];  
    for (int i = 0; i < arr.length; i++) {  
        result[i] = arr[arr.length - i];  
    }  
    return result;  
}
```